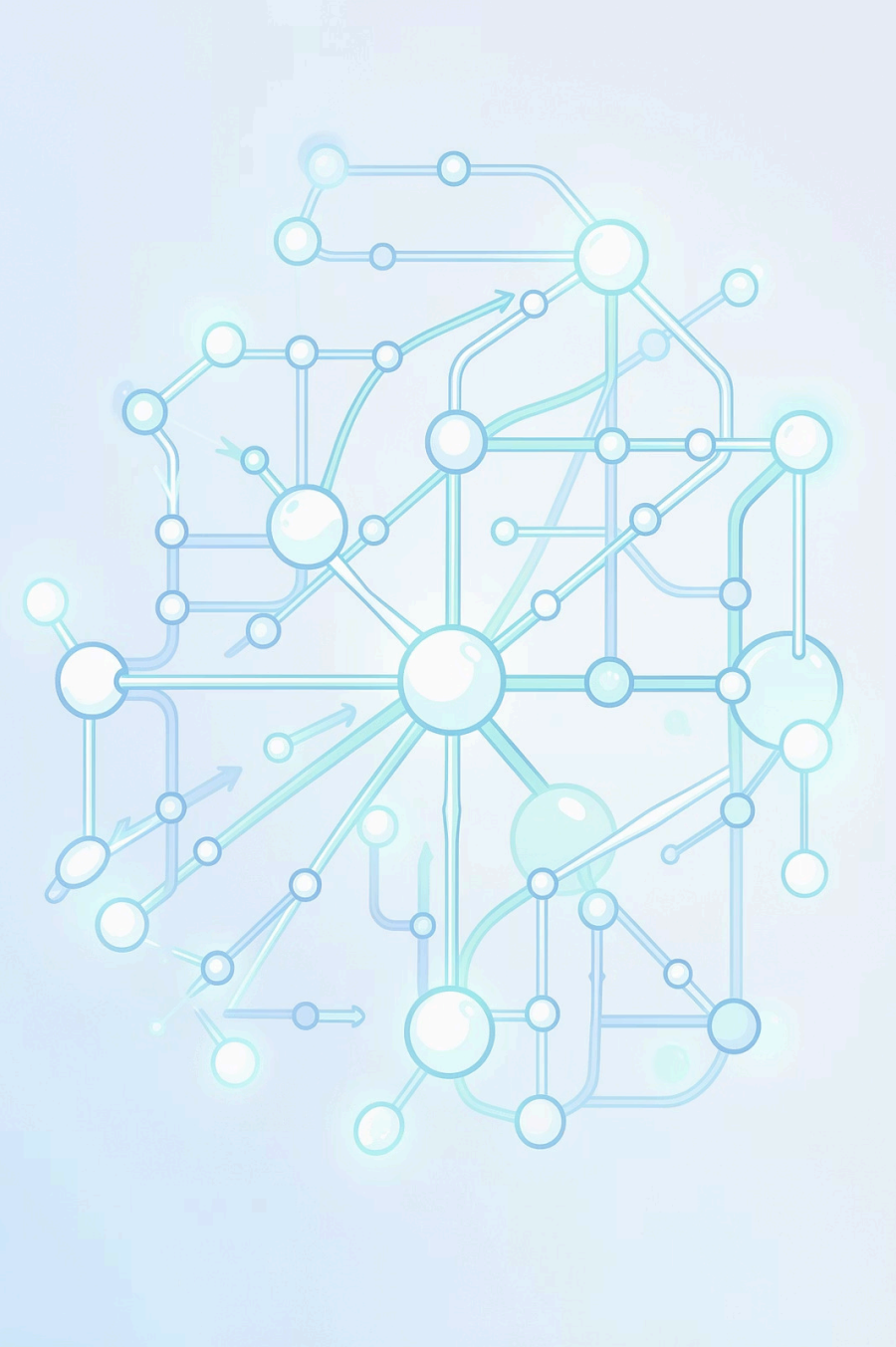




Algorithms: The Secret Sauce of Problem Solving

Every app, search engine, and smart device runs on one fundamental idea – a step-by-step plan to get things done. Welcome to the world of algorithms.



FOUNDATION

What Is an Algorithm?

An algorithm is a **step-by-step procedure or set of rules** for solving a specific problem. Think of it as a recipe: precise instructions that, when followed correctly, always produce the same result.

Input

Data to work with

Process

Steps to execute

Output

The solution

REAL WORLD

Algorithms in Everyday Life

You already use algorithms – you just don't call them that. Any time you follow a sequence of steps to reach a goal, you're thinking algorithmically.



Driving Directions

GPS calculates the fastest route using a routing algorithm.



Adding Numbers

Even basic arithmetic follows a defined, repeatable procedure.



Assembling Furniture

IKEA instructions are a physical algorithm – step 1 through step N.

Why Are Algorithms Important?

Clarity

They eliminate ambiguity – every step is explicit and unambiguous.

Repeatability

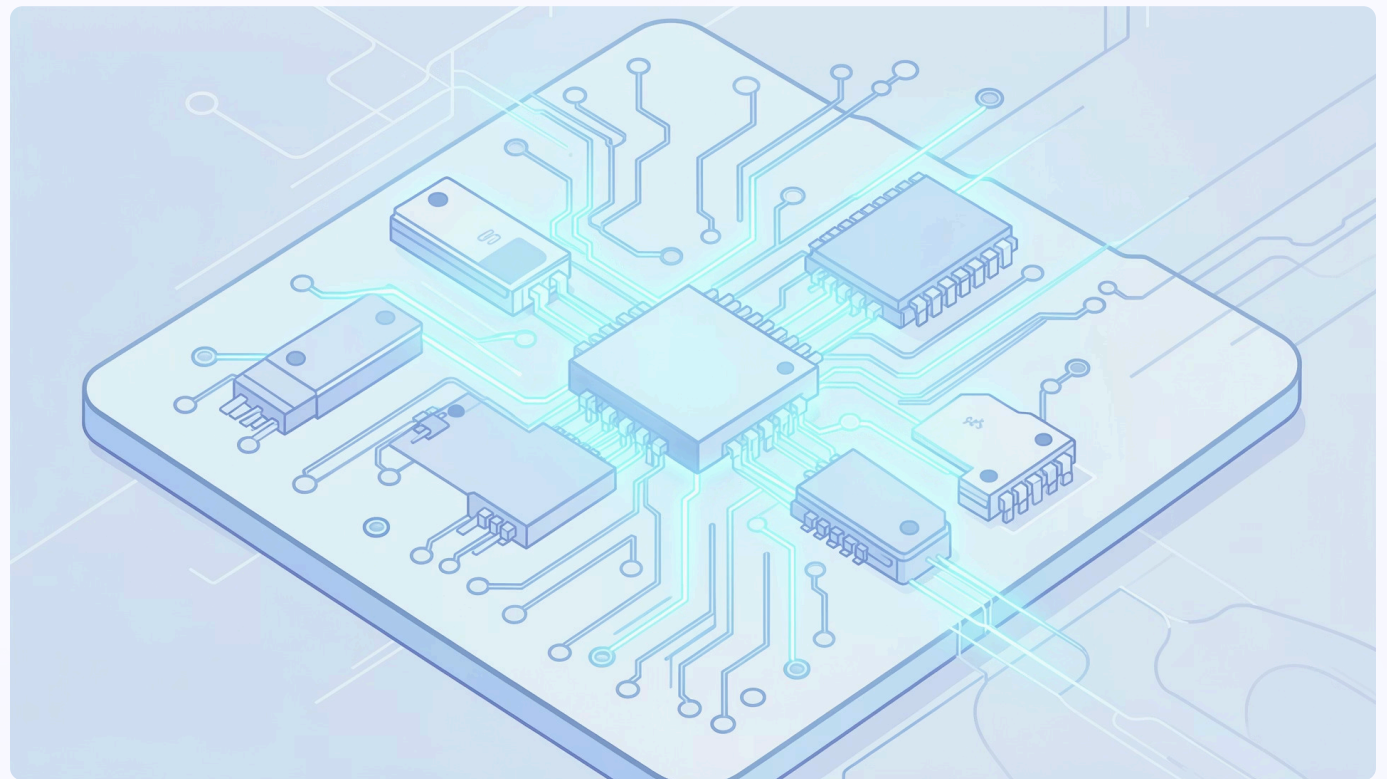
The same input always produces the same output, every single time.

Efficiency

Good algorithms solve problems faster and with fewer resources.

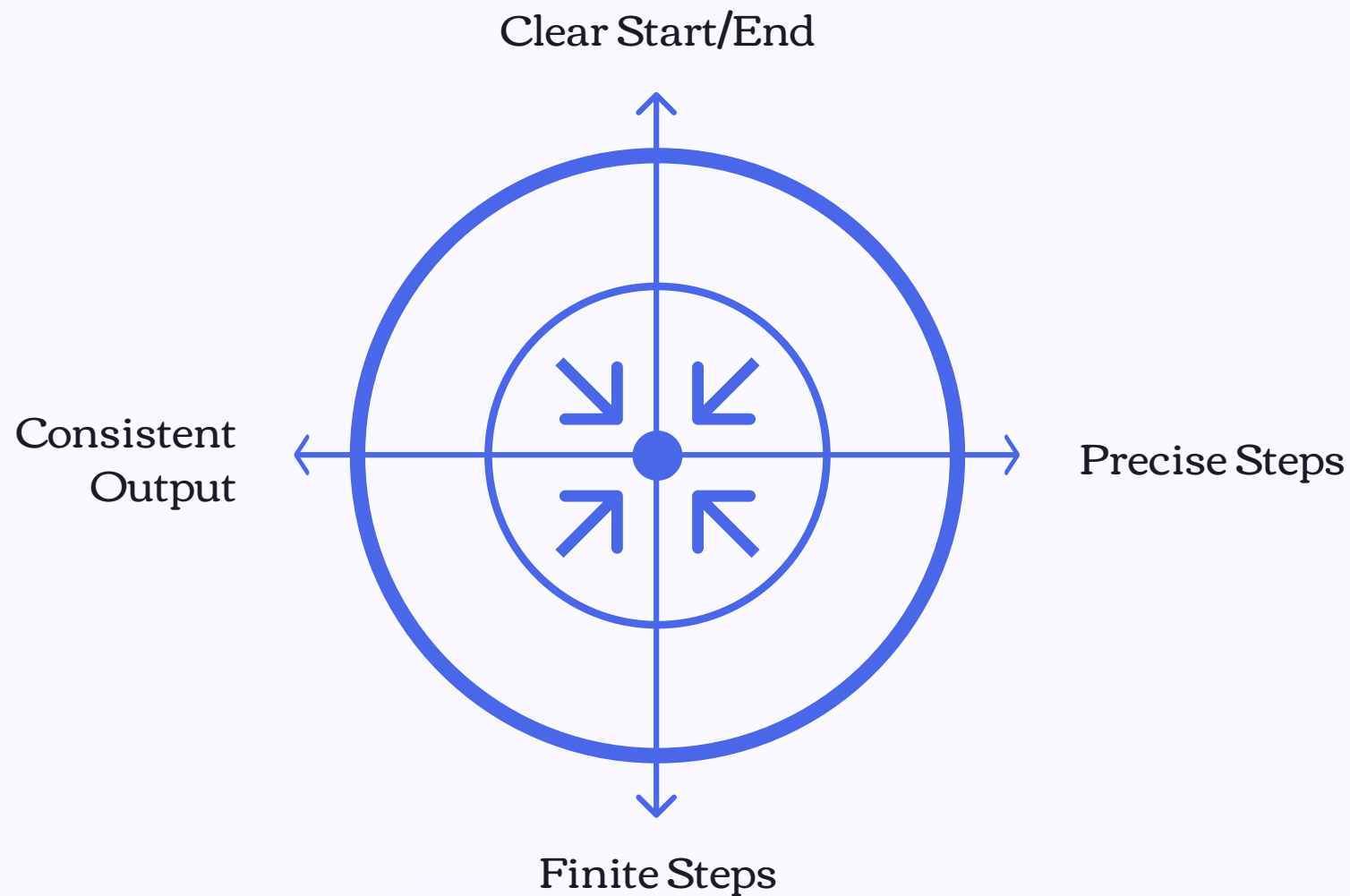
Foundation of Computing

Computers cannot act without a precise set of instructions to follow.

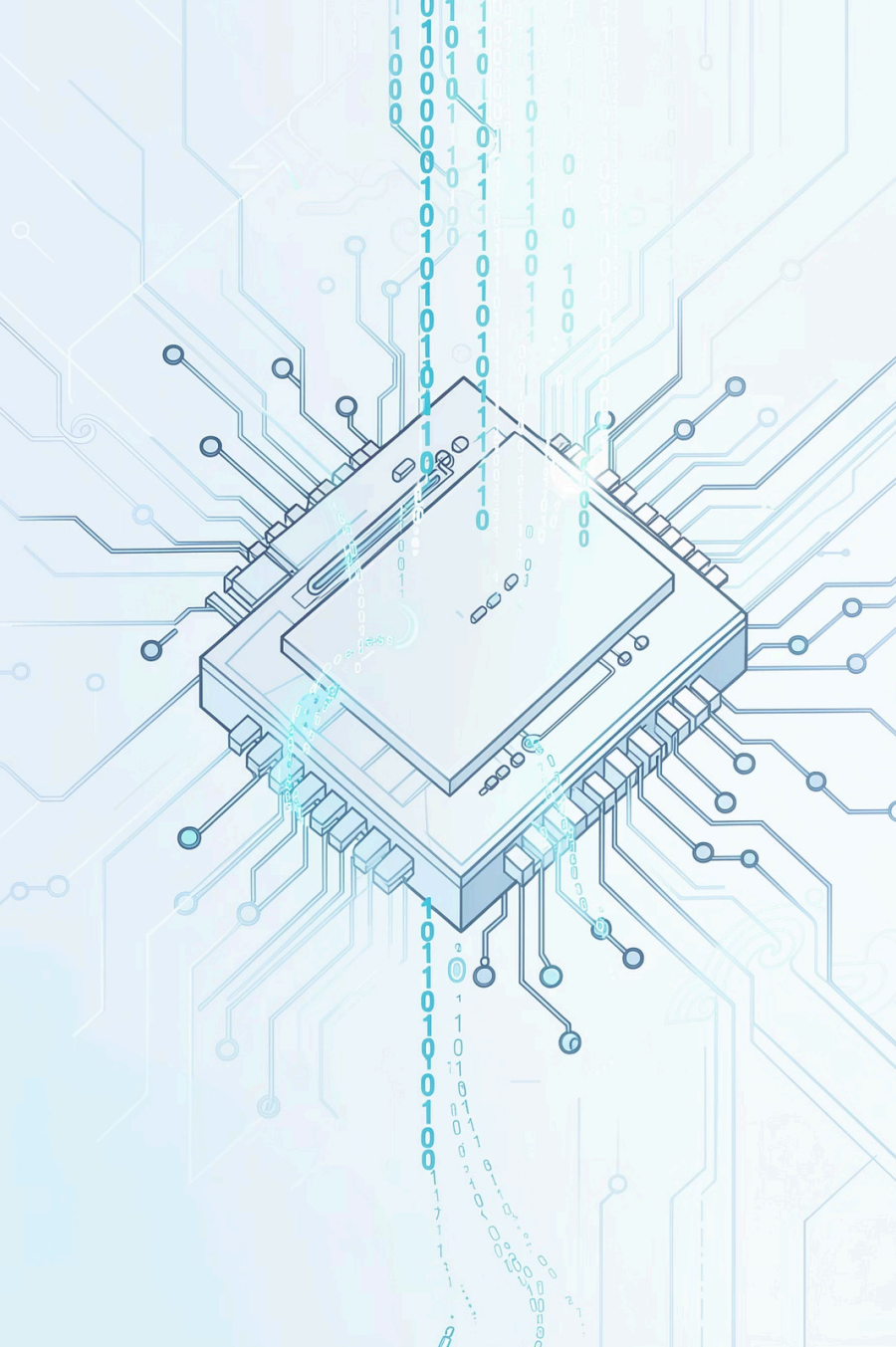


The Anatomy of an Algorithm

Not every list of steps qualifies as an algorithm. A proper algorithm must satisfy specific structural requirements to be valid and executable.



These four properties ensure that an algorithm is reliable, predictable, and useful – whether executed by a human or a machine.



COMPUTING

Computers Need Algorithms

Unlike humans, computers have **no intuition**. They can only make binary yes/no decisions and follow exact, pre-defined instructions. Every task – from playing a video to launching a rocket – must first be broken down into an algorithm.

Human Thinking

Intuitive, flexible, handles ambiguity

Computer Thinking

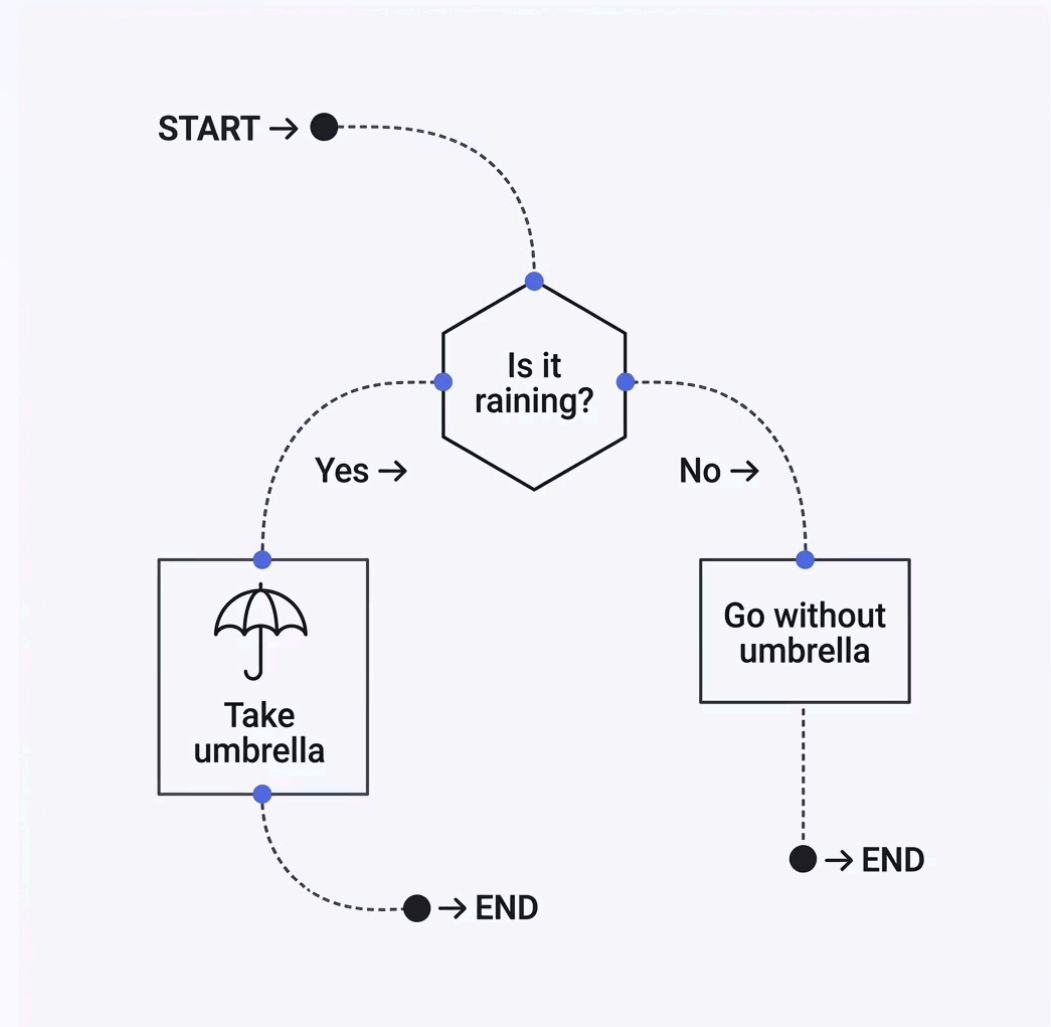
Literal, precise, needs exact instructions

The Bridge

Algorithms translate human goals into machine-executable steps

Control Flow: The Path of Decisions

Control flow determines **which steps execute and in what order**. Decision points (like "Is it raining?") branch the algorithm into different paths, allowing it to respond dynamically to conditions.



→ Sequence

Execute steps one after another in order.

→ Selection

Choose a path based on a condition (if/else).

→ Iteration

Repeat a step until a condition is met (loops).

What Computers Can't Do (Yet!)



Algorithms are powerful, but they have boundaries. Computers lack the **intuition, context, and lived experience** that humans bring to complex, subjective problems.

Subjective Judgement

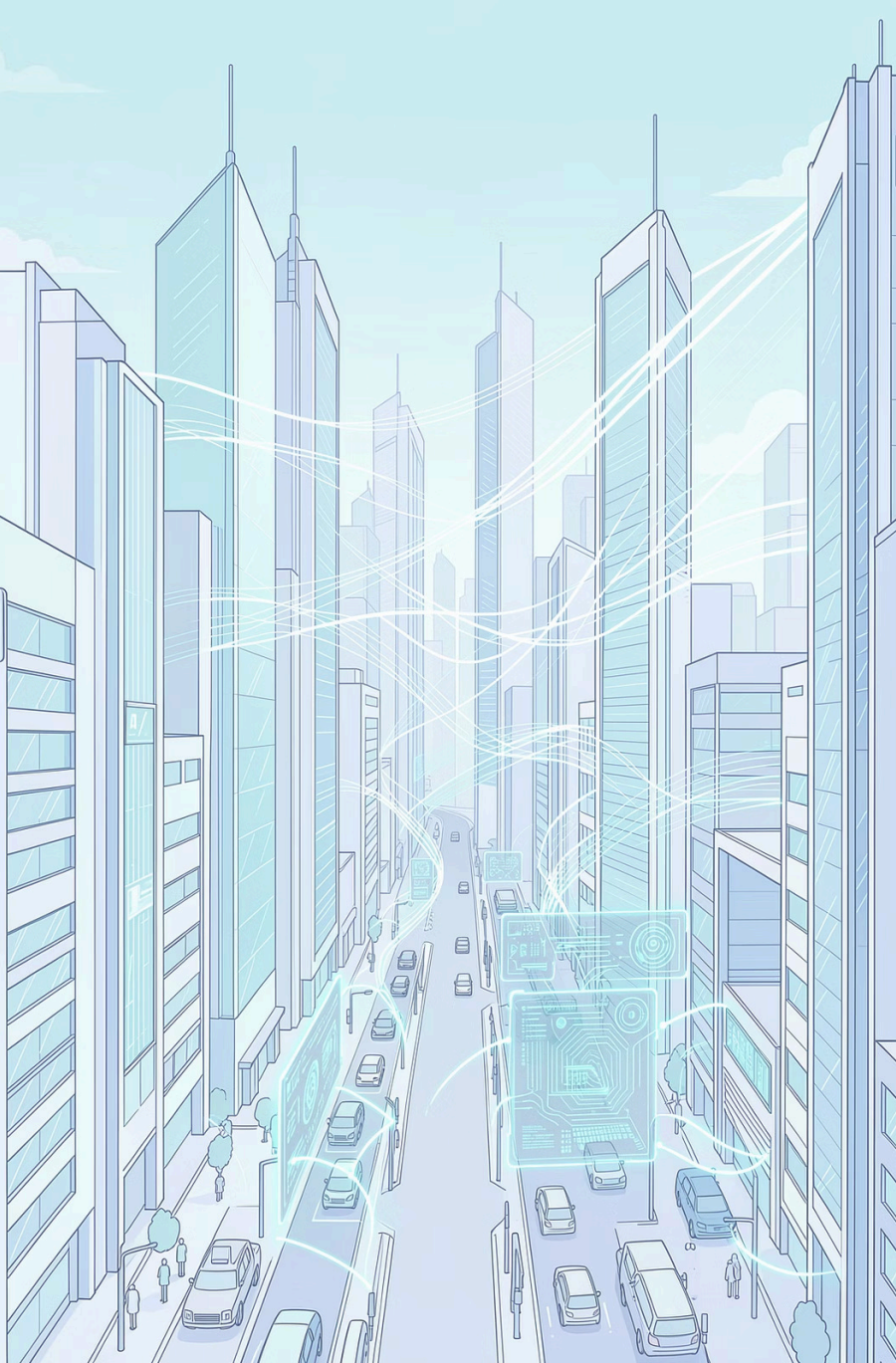
"Is this website useful?" requires human context and intent.

Emotional Nuance

"Does this person dislike me?" involves reading tone, body language, and history.

Creative Ambiguity

Art, humor, and cultural meaning resist clean algorithmic rules.



IMPACT

The Power of Algorithms

From sorting your emails to training AI models, algorithms are the **backbone of modern technology**. They enable complex tasks to be performed efficiently, accurately, and at scale.

4.9B

Searches Per Day

Powered by ranking algorithms

500M

Tweets Per Day

Filtered and sorted in real time

1B+

Maps Routes

Calculated daily by GPS algorithms

Made with GAMMA

TAKEAWAY

Algorithms: Your Toolkit for the Future

Understanding algorithms means understanding **how technology thinks**. Whether you're coding, analyzing data, or simply navigating a complex world, algorithmic thinking is your most powerful problem-solving superpower.

Think Clearly

Break big problems into small, manageable steps.

Build Precisely

Write instructions that are unambiguous and repeatable.

Solve Anything

From sorting lists to training AI – the logic is the same.

